Thesis Report

June 13, 2026

Summary

The thesis is technically ambitious and generally well organised. The main exposition is clear, and the finite-sample chapter is unusually transparent about limitations. The main issues to address before submission are: (i) excessive repetition of “source”, “route”, “framework”, and “this work”; (ii) some notation and terminology that can mislead readers; (iii) front-matter/layout problems; and (iv) interval-calibration claims that need sharper qualification.

1 High-priority technical and structural comments

1. **Add a short thesis roadmap of assumptions.** The reader has to track many conditions: first order, second order, finite mean, $\rho < 0$, bridge-rate condition, source-weight consistency, plug-in consistency, and iid common margins. A table at the end of Chapter 2 or start of Chapter 3 would greatly improve readability.

2. **Strengthen the contribution statement.** The thesis says it “formulates and supports” first-order limits. This is honest, but the contribution would be clearer if split into: estimator definition; exact decomposition; conditional first-order transfer; finite-sample diagnostics. This would also make the conjecture/theorem status transparent.

3. **Be more precise about finite-sample inference.** Chapter 4 correctly reports undercoverage, but the text should state earlier that the confidence intervals are asymptotic diagnostics and should not be presented as calibrated finite-sample procedures.

2 Detailed list of suggested corrections

PDF p.	Line	Current text/location	Issue	Suggested correction
1–4	–	Blank pages before the table of contents	The PDF begins with four blank pages. This looks like a compilation/front-matter error unless intentionally anonymised.	Add title page, abstract, acknowledgements, and declaration if required; otherwise remove blank pages.
5–6	–	Contents starts on PDF p. 5 and printed p. 1	Front matter is missing or suppressed. For an undergraduate thesis, this may look incomplete.	Add a proper title page and abstract, and use roman numerals for front matter if required by the programme.
7	15–19	“Expectiles, introduced by...”	The paragraph mixes definition, risk-measure properties, and asymptotic tail theory before the reader has seen the formal definition.	Add one transition sentence: “A formal definition is recalled in Section 2.3; here I only summarise why high-level expectiles are relevant for tail risk.”

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8	15–21	Estimator displayed with τ_n but later local Weissman estimators are written at $1 - p_n$	The notation is correct because $\tau_n = 1 - p_n$, but using both forms can distract.	Write $\widehat{q}_n^*(1 - p_n \mid \omega)$ consistently, or explicitly say "we write $\tau_n = 1 - p_n$ and use the two notations interchangeably."
8	31	"finite-m iid common-marginal setting"	Informal mathematical typography.	Use "finite-m i.i.d. common-marginal setting".
9	7–12	"This work reports... Their role is..."	The contribution paragraph is clear but repetitive: "this work" appears repeatedly in the introduction.	Vary phrasing: "The third contribution is a finite-sample study... The simulations are designed to assess..."
10	10	"survival function $F = 1 - F$"	Notational collision: the survival function should not be denoted by the same F as the distribution function. This is a substantive notation problem.	Use $\overline{F} = 1 - F$ consistently. In the source, check that the command for the bar has not been dropped.
10	15–23	Definition of $U(t)$ using $1/F(x)$	This depends on the survival function, but because the survival notation is unclear it becomes ambiguous.	Write $U(t) = F^{\leftarrow}(1 - 1/t)$ for continuous F, or $U(t) = \inf\{x : 1/\overline{F}(x) \geq t\}$.
11	26	"F-distribution tails"	Potential confusion with the distribution function F.	Use "Fisher–Snedecor F distribution tails" or "F-law tails".
12	26–32	Theorem 2.5 paragraph	The theorem states the Hill CLT, but the endpoint $\rho = 0$ is mentioned through another source. This is unusual and should be justified more carefully.	Add one sentence explaining exactly why the cited Daouia et al. result covers the endpoint in the $m = 1$ case.
13	1–2	"k controls... fixed outside the asymptotic analysis"	Since $k = k_n \to \infty$, saying it is "fixed" is misleading.	Replace with "The threshold sequence k_n is treated as an exogenous tuning sequence in the asymptotic analysis."
13	23–25	"the case $np = O(1)$ is very-extreme..."	Useful comment, but it would benefit from stating whether the theorem covers $np = O(1)$ under the displayed assumptions.	Add: "This is included when $k/(np) \to \infty$ and $\sqrt{k}/\log(k/(np)) \to \infty$ hold."
15	11–15	"coherent exactly when $\tau \geq 1/2$"	This is a strong statement. It should be tied carefully to the convention "losses" vs. "positions".	Add a sentence clarifying the sign convention and whether the statement concerns $\rho_\tau(Y) = \xi_\tau(-Y)$ or $\xi_\tau(X)$ directly.
16	34–41	Second-order bridge expansion	The expression is central but visually dense.	Consider isolating the A-bias and mean-bias terms in separate displayed equations, then define $c(\gamma, \rho)$ below.
17	5–12	"Relation to earlier expectile estimators"	The section is informative but too brief for a literature subsection.	Add one or two sentences contrasting the proposed estimator with direct empirical expectiles, LAWS-type estimators, and intermediate-expectile extrapolation.

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18	8–14	Setup for m samples	The text switches from distributed iid later to general heterogeneous margins here.	Add a signpost: "This general setup is recalled only to import notation; Chapter 3 specialises to common iid margins."
19	21–33	Theorem 2.9 covariance matrix	Very notation-heavy; the reader may lose the special case used later.	After the theorem, add a short explanation of b_j, c_j, B_c, and V_c in words.
22	21–28	Theorem 2.12 assumptions	The theorem is quoted, but its dependence on tail homoskedasticity and $\rho_j < 0$ deserves emphasis because Chapter 3 relies on it.	Add "These two assumptions are essential for transferring the extreme-quantile result to a common target."
24	21–33	Proposition 2.14 and source consistency route	Good content, but the paragraph is long and condition-heavy.	Convert the consistency assumptions into a bullet list. This will make later references to "source-admissible" clearer.
25	18–21	Distributed special case bullet list	Good material, but it appears late.	Consider moving this before the fully general formulae, or add a boxed "case used in Chapter 3".
26	7–17	Opening of Chapter 3	Strong paragraph, but "The role of the chapter is to show" is informal.	Replace with "This chapter identifies, via an exact log decomposition, which terms remain visible..."
27	1–14	Estimator definition	The notation $\widehat{\xi}^{\mathrm{pool},*}_{\tau_n}(\nu, \omega)$ is central but not visually emphasised.	Add a short sentence after (3.1): "All subsequent distributed estimators are special cases of (3.1)."
28	43	"The following conjecture records..."	If the result is proved in the subsequent argument, calling it a conjecture weakens the thesis.	Decide: rename as "Theorem 3.2" if complete; otherwise state the exact missing proof ingredient.
29	31–39	Bridge-rate discussion	The example is useful but compressed.	Expand slightly: explain why $1 - p_n \geq u_n$ means the target quantile is no smaller than the intermediate anchor quantile.
30	37–44	"All logarithmic and Taylor expansions..."	This compact-event paragraph is important and proof-like.	If keeping the result as a theorem, split this into a formal lemma verifying the high-probability event.
32	1–11	Local assumptions for Theorem 2.9	The derivation is good, but line-by-line notation is hard to follow.	Add a concluding sentence: "Thus each local Hill estimator has the required $\sqrt{k_j}$ second-order centring, and the vector CLT applies."
34	35–46	Remark 3.3	Clear and important.	Consider moving the first sentence into the theorem statement, because it is one of the main conceptual messages.
35	20	"Conjecture 3.4"	This is not really a conjecture; it is an algebraic consequence of Conjecture 3.2.	Rename it "Corollary 3.4" conditional on Theorem/Conjecture 3.2.
37	21–31	"The deterministic conjecture is the base statement..."	The word "source" is overused and undefined for a new reader.	Define "source" once as "the Daouia et al. (2024) pooling theory" and then use that phrase sparingly.

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39	9–30	Conjecture 3.7 interval	The interval formula is important but hard to parse in prose.	Add a short derivation line before the formula: "Solving the studentised inequality for ξ_{τ_n} gives..."
40	39–47	Remark 3.8	Good qualification, but it would be more useful before the simulation chapter.	Cross-reference this remark at the start of Section 4.5.
41	7–19	Opening of Chapter 4	The paragraph states the simulation scope well.	Add the number of replications and the main target levels here, not only in Section 4.2.
41	30–31	"intervals under-cover"	Hyphenation is nonstandard.	Use "undercover" as a verb? Better: "intervals have empirical coverage below the nominal level".
42	13–15	"DPS is used as a compact label..."	Acronym "DPS" is introduced late and may feel informal.	Define it in Chapter 2 or avoid the acronym in headings/tables if space permits.
42	26–27	"estimated using evt0 (Belzile et al. 2026)"	Package citation is fine, but future/online package references should be checked for final submission date.	Verify version/date and use a standard software citation format; avoid future-looking access dates unless submission is in 2026.
43	19–25	Simulation grid description	Several design parameters are packed into one paragraph.	Present a compact design table: DGPs, n, m, k_j/n_j, np, allocation, replications.
45	20–26	Explanation of $d_j = 1$ and AMSE collapse	This is one of the best explanations in the chapter.	Move a shorter version before Table 4.1 so the reader understands why AMSE and variance rows coincide.
47	Figure 4.1 caption	Caption repeats too much of the surrounding text.	Shorten the caption and move interpretation to the main text.	
50	6–14	Section 4.5 interval interpretation	The centralised benchmark undercoverage is important.	Add a more explicit warning: "Therefore these intervals should not be recommended without additional calibration in samples of this size."
51	Table 4.4	Exact-Pareto distributed rows omitted, centralised rows shown	The convention is explained, but readers may misread dashes as failures.	Add a table footnote: "– indicates source-ineligible plug-in route, not zero coverage or numerical failure."
52	33	"colour records"	British spelling is fine if used consistently, but the rest of the text mostly uses American technical style.	Use either "colour" throughout or "color" throughout; consistency matters.
53	10–21	Decomposition diagnostics	Strong analysis, but A_n, B_n, and C_n should be redefined briefly for readers who jump to Chapter 4.	Add a parenthetical reminder: "where A_n is the pooled-Weissman term, B_n the bridge-Hill term, and C_n the population bridge gap."
54	15–25	Threshold sensitivity discussion	Good, but it would benefit from a practical conclusion.	Add one sentence: "This suggests reporting threshold-sensitivity plots as part of any empirical application."

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57	29–32	"first-order plug-in intervals may under-cover..."	Good conclusion, but slightly repetitive with Chapter 4.	Condense and refer to the decomposition diagnostics rather than restating all mechanisms.
58	1–3	"does not address non-identical margins..."	Excellent limitation statement.	Move a shorter version to the end of the introduction so the scope is clear from the beginning.
60	10	"Looking back"	Informal phrase for a conclusion.	Replace with "The two-weight formulation separates two roles that are otherwise easy to conflate..."
61	33–35	"The first-order log-scale intervals under-cover..."	Same hyphenation issue as above.	Replace with "have empirical coverage below nominal levels".
62	26–32	"DPS weight theory transfer"	Acronym and "transfer" wording are repeated many times.	Use "the Daouia et al. pooling weights apply to ω at first order" for clarity.
63	28–33	Alternative architectures	Good research direction, but it could be more concrete.	Add examples of what extra summaries would be communicated: local upper order statistics, local tail means, intermediate expectiles, or local estimates of $E[X]$.
66	9–15	Software references dated 2026	The bibliography includes future package versions/access dates relative to many submission contexts.	Verify the thesis submission date. If submitted before 2026-06-02, update these references to actual available versions.

3 Writing-style recommendations

1. **Reduce repetitive framing.** Frequent phrases include "this work", "route", "source", "source-admissible", "framework", and "first-order". Keep the technical terms when needed, but replace generic repetitions with direct subjects.

2. **Use shorter sentences in proof-heavy passages.** Several proof paragraphs contain multiple assumptions, conclusions, and references in one sentence. Split them into assumption, implication, and conclusion.

3. **Prefer theorem/proposition/corollary labels to conjecture labels when proofs are supplied.** The current labelling may make the thesis seem less rigorous than it is.

4. **Add more explanatory bridges between chapters.** Chapter 2 is mathematically dense. A one-page "how the ingredients are used later" summary would make Chapter 3 easier to read.

5. **Standardise terminology.** Choose one form: "plug-in" or "plugin", "undercoverage" or "coverage below nominal", "i.i.d." or "iid", "color" or "colour".

4 Research-development suggestions

1. **Resolve the theorem/conjecture status.** This is the main research issue. If the supporting arguments are complete, formalise them as theorems. If not, isolate the missing technical

result and state it as an assumption or lemma-to-be-proved.

2. **Add a lower-order analysis of the bridge terms.** The simulations show that $B_n(\nu)$ and C_n can be numerically visible even when asymptotically negligible. A lower-order expansion would connect theory and finite-sample evidence more convincingly.

3. **Investigate bridge-Hill weights ν.** The thesis correctly says ν is first-order unidentified. A natural extension is to minimise an approximate lower-order MSE involving $m(\gamma)\{\widehat{\gamma}_n(\nu) - \gamma\}$.

4. **Improve interval calibration.** Since both distributed and centralised intervals undercover in several settings, consider bias-corrected centring, bootstrap/subsampling diagnostics, or intervals that retain the population bridge term where estimable.

5. **Add a real-data illustration if feasible.** Even a short application would clarify why distributed extreme expectiles are useful in practice and what communication constraints motivate the estimator.

6. **Report threshold sensitivity graphically.** Tables are comprehensive but hard to digest. Plots of log RMSE and coverage versus k/n would communicate the threshold trade-off more effectively.

Final assessment

The thesis has a solid conceptual core: the two-weight decomposition is a useful way to understand which part of the pooled extreme-expectile estimator inherits the Daouia et al. pooling theory. The strongest revision would be to sharpen the formal mathematical status of Chapter 3 and to polish the writing so that the technical contribution is easier to see.